Fraud Assignment

You are in charge of developing a framework to detect fraud for a life insurance company. The life insurance company services many the entire nation, but you will only be working with customer data from their west coast branches. The company has never done any fraud modeling in the past, so you will be starting from scratch.

Anomaly Detection

One of the problems when building a fraud detection system from scratch is not knowing any fraudulent claims to try and build a model from. In this situation you must use unsupervised learning or anomaly detection techniques followed by domain expertise to try and evaluate where fraud might exist. Your task is to identify groups of observations (customers in this case) you are suspicious of fraudulent acts. You will submit these groups of observations. A “domain expert” will look through your groups and determine if those groups exhibit fraudulent behavior. The group that can identify the most actual fraudulent cases (I know the true flags of each observation) will have their group’s anomalies used for modeling in the subsequent stages.

Submit no more than 2000 observations that you believe are potentially fraudulent. Email a list of Customer ID’s in a CSV file to sjsimmo2@ncsu.edu by the **due date of March 24 at 5:00pm**.

Data Dictionary – MSA2026

Name of Variable	Variable Description
GivenName	First name.
MiddleInitial	Middle name initial (if applicable).
Surname	Last name.
Street Address	Street address for customer.
City	City for customer.
State	State for customer.
ZipCode	ZIP code for customer.
Country	Country name for customer.
TelephoneNumber	Last reported telephone number for customer.
MothersMaiden	Mother's maiden surname.
Birthday	Date of birth for the customer.
BloodType	Blood type for the customer.
Pounds	Weight in pounds of the customer.
FeetInches	Height in feet and inches of the customer.
Date_Initial	Date of initial opening of the policy.
Cust_ID	Customer ID.
Cov_ID	ID for the specific policy number. Customers may have multiple policies.
Type	Type of policy with the following levels: • T (Term life insurance) • W (Whole life insurance) • V (Variable life insurance)
Reward_Reason	Reason for reward by type of death code: • 100-199 (Accidental Death) • 200-299 (Criminal Acts) • 300-499 (Health Related Causes) • 500-549 (Dangerous Activity - Exclusion) • 550-559 (War - Exclusion) • 560-569 (Aviation – Exclusion) 570-579 (Suicide – Exclusion)
Reward_Amount	Amount rewarded. Missing value if no reward on that transaction.
Tobacco_IN	Does the customer use tobacco (Y or N). Filled out on initial questionnaire.
Caffiene_IN	Does the customer use caffiene (Y or N). Filled out on initial questionnaire.
Alcohol_IN	Does the customer use alcohol (Y or N). Filled out on initial questionnaire.
Number_Changes	Number of changes in coverage limit over lifetime of policy
Cov_Limit_Claim	Coverage limit at time of claim.
Income_Claim	Income at time of claim.
Adj_ZIP	The ZIP code of the adjuster handling the transaction.
Adj_ID	ID of the adjuster.
Tech_ID	ID of the technician working with adjuster on claim.

Adj_Tech_Confidence	Confidence metric from association analysis between Adjuster and Technician
Tech_Adj_Confidence	Confidence metric from association analysis between Technician and Adjuster
Initial_to_Claim_Difference_CL	Difference in coverage limit from initial policy amount to claim amount.
Distance_Claim_Adj	Distance (in miles) from claim ZIP to adjuster ZIP.
Coverage_Income_Ratio_Claim	Coverage to income ratio at time of claim.
Coverage_Income_Ratio_Initial	Coverage to income ratio at initial policy creation.
Time_LastChange_Claim	Time between last change and claim.
Cov_Limit_Slope	Slope of Coverage limit across whole dataset
Marriage	Married, Single, Widowed
CH_Count_6Mon	Number of coverage changes in last six months before claim.
Med_BP_Flag	Flag for blood pressure history.
Med_Chol_Flag	Flag for cholesterol history.
Med_Arth_Flag	Flag for arthritis history.
Med_Can_Flag	Flag for cancer history.
Med_Diab_Flag	Flag for diabetes history.
Med_Asth_Flag	Flag for asthma history.
Med_Gla_Flag	Flag for glacoma history.
Med_Kid_Flag	Flag for kidney disease history.
Med_Leuk_Flag	Flag for leukemia history.
Med_Ment_Flag	Flag for mental illness history.
Med_HA_Flag	Flag for heart attack history.
Med_SE_Flag	Flag for seizures history.
Med_SCA_Flag	Flag for sudden cardiac arrest history.
Med_Str_Flag	Flag for stroke history.
Med_TB_Flag	Flag for tuberculosis history.
Med_UI_Flag	Flag for urinary infection history.
MEDICAL HISTORY	Same variables exist in the dataset for both paternal father and maternal mother of the client
Susp_Adj_Tech	Flag for adjuster technician combo that is suspicious based off network analysis.
Changes_Med_History	Number of changes of medical history.
Med_Deletion_Flag	Flag for whether a medical history was previous present, but subsequently deleted.
Time_Between_CL_R	Time (in days) between claim and reward.
Time_Between_IN_CL	Time (in days) between initial and claim.
Med_Conditions	Number of medical conditions the person had
Exclusion	Flag for whether a policy was an exclusion reward reason.
Pop_CZIP	The census population for the customer’s zipcode.
MedInc_CZIP	The census median income for the customer’s zipcode.
Pol_Count	Number of policies for the customer.
Exclusion_Payout	Flag for whether the exclusion reason was present but a payout was issued anyway.

Ratio_CL_PopZIP	Ratio between number of claims in ZIP to population (in thousands) of the ZIP
ZScore_Inc_Change	Z-score of change in income compared to the change in income across everyone
Ratio_Cov_Limit_Rew	Ratio of coverage limit to the reward amount
CH_per_Year	Number of changes per year
Time_Between_CH_R	Time (in days) between last change and reward
PerSize_Last_CH	Percentage change in the last change in coverage limit (0 if no change)
Per_R	Percentage of coverage limit that was paid out
Per_R_CL_by_Adj	Percentage of coverage limit that was paid out on average for that specific adjuster
R_Greater_CL	Reward greater than claim (0 if no, 1 if yes)
Ratio_CH_TimeCompany	Ratio of number of changes to time at company (in years)
CL_per_Adj	Number of claims processed by that adjuster
Diff_Cust_Inc_CZIPInc	Difference between customer income (at claim) to the median income of the customer's ZIP
Avg_CH_CovLimit	Average value of the changes in coverage limit (0 if none)
IN_LessThan_60	If policy was greater within 60 days of claim (0 if no, 1 if yes)
Ratio_AdjZIP_PopZIP	Ratio of number of adjusters per ZIP to population of ZIP
Num_Adj_CustID	Number of adjusters for that specific customer ID
Zip_per_CustID	Number of ZIP per customer ID
Avg_CL	Average coverage limit amount across customer duration
PerSize_Max_CH	Percentage maximum change in coverage limit
SD_CovLim_Last_6mon	Standard deviation of the coverage limit over the last six months (0 if no change)
Time_MaritalCH_CL	Time between last marital status and claim